



 **Time:** 5 minutes

 **Outcome:** Calculate Total Surplus & Gains from Exchange.

Difficulty: ●●○ Intermediate

Total Surplus & Gains from Exchange



THE CORE IDEA

Total surplus equals consumer surplus plus producer surplus and measures the gains created by exchange. Each trade adds to total surplus when the buyer's willingness to pay exceeds the seller's opportunity cost. Total surplus is maximized when every mutually beneficial trade occurs and no trade with a cost above its value is completed.



HOW TO RECOGNIZE IT

- Add consumer surplus and producer surplus to obtain total surplus.
- For each possible trade, compare the buyer's willingness to pay with the seller's opportunity cost.
- A trade creates gains when buyer value exceeds seller cost.
- On a graph, total surplus is the area between demand and supply up to the quantity traded.

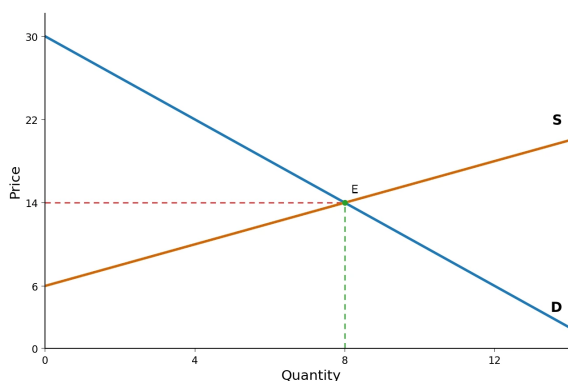


WATCH OUT

A common mistake is to fail to separate transfer from total loss. Consumer surplus is \$10 and producer surplus is \$8; their sum equals the \$18 gain from exchange.



WORKED EXAMPLE: CALCULATING TOTAL SURPLUS



The graph's equilibrium is 8 units at \$14. Consumer surplus is $\frac{1}{2} \times 8 \times (\$30 - \$14) = \64 , and producer surplus is $\frac{1}{2} \times 8 \times (\$14 - \$6) = \32 . Total surplus is therefore $\$64 + \$32 = \$96$.



CHECK YOURSELF

Buyer value is \$17 and seller cost is \$22. Efficient action?



READY?

Return to the game and master this concept.